Review plan · for the lecturer

# Teaching cyber security at university: unit check

Cyber security · University · Unit check

| Time | Materials |
| --- | --- |
| 40 min | Projector or board, and the lecture slides. |

## Learning goals

- Principles and STRIDE
- Public-key cryptography
- Password storage and MFA
- Ethics and assessment

## Activity: Peer-instruction review

1. Vote (10 min): Show each question; students answer alone and commit to an answer.
2. Discuss (15 min): Where answers split, pairs argue for two minutes and vote again.
3. Explain (10 min): Work through the questions most students missed, showing the reasoning step by step.
4. Plan (5 min): Note which misconception to revisit next week and how.

## Discussion

- Which misconception will you target first?
- Which lab will you add?
- How will you change CTF assessment?

## Exit ticket

- One change to my security course this term, and how I will know it worked.

**Try it in your next lecture**
Bring your rules of engagement and CTF rubric to your next department meeting for review.

Quiz sheet · for students

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Name _____ Group _____ Date _____

**1** **The portal checks a student's permission at sign-in but not on later requests for other students' grades. The principle broken is**

◯ Open design ◯ Psychological acceptability ◯ Complete mediation

**2** **Match each STRIDE threat to the property it violates.**

| | | |
|---|---|---|
| Spoofing | • • • • | Confidentiality |
| Information disclosure | • • • • | Availability |
| Denial of service | • • • • | Authentication |

**3** **Compute a public value.**

With p = 23, g = 5 and b = 2, Bob sends B = _____ .

**Word bank:** 10 · 25 · 2

**4** **Encrypting a message with Bob's public key proves who sent it.**

◯ True ◯ False

**5** **Plain Diffie–Hellman is vulnerable to**

◯ Collisions ◯ Rainbow tables ◯ Man-in-the-middle

**6** **Which make leaked password tables harder to crack? Choose all that apply.**

☐ Argon2id ☐ Unique salts ☐ Faster hashing

**7** **The factor that resists real-time phishing is**

◯ A FIDO2 hardware key ◯ An SMS code ◯ A security question

**8** **Lab VMs for attack practice should use a bridged network so students can reach real targets.**

◯ True ◯ False

**9** **Order the safe offensive-lab sequence.**

☐ Isolated practice
☐ Graded write-up
☐ Law and ethics
☐ Signed scope

**10** **Two students submit the same flag a minute apart; only one write-up explains the exploit and the fix. Their marks should rest on**

◯ The flag alone, since both found it ◯ Who submitted the flag first
◯ The write-ups and a short oral check

Answers · for the lecturer

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1 **Complete mediation**

Complete mediation: check every access, not just the first.

2 **Spoofing — Authentication**
**Information disclosure — Confidentiality**
**Denial of service — Availability**

Each letter targets one property.

3 **With p = 23, g = 5 and b = 2, Bob sends B = 2.**

25 mod 23 = 2.

4 **False**

Anyone can do that; sender proof needs a signature.

5 **Man-in-the-middle**

It authenticates no one.

6 **Unique salts; Argon2id**

Speed helps the attacker.

7 **A FIDO2 hardware key**

It is bound to the real origin.

8 **False**

Use host-only isolation.

9 **Top to bottom: 3, 4, 1, 2**

Consent before practice.

10 **The write-ups and a short oral check**

Flags can be shared; the write-up and an oral check show who understood the exploit.